INCH-POUND

MIL-DTL-28840/14G
w/AMENDMENT 1
16 May 2019
SUPERSEDING
MIL-DTL-28840/14G
14 November 2013

DETAIL SPECIFICATION SHEET

CONNECTORS, ELECTRICAL, CIRCULAR, THREADED, HIGH SHOCK, HIGH DENSITY, SHIPBOARD, CRIMP CONTACTS, RECEPTACLE, JAM NUT MOUNTING, CLASSES D, DS, P, T AND Z

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-28840.

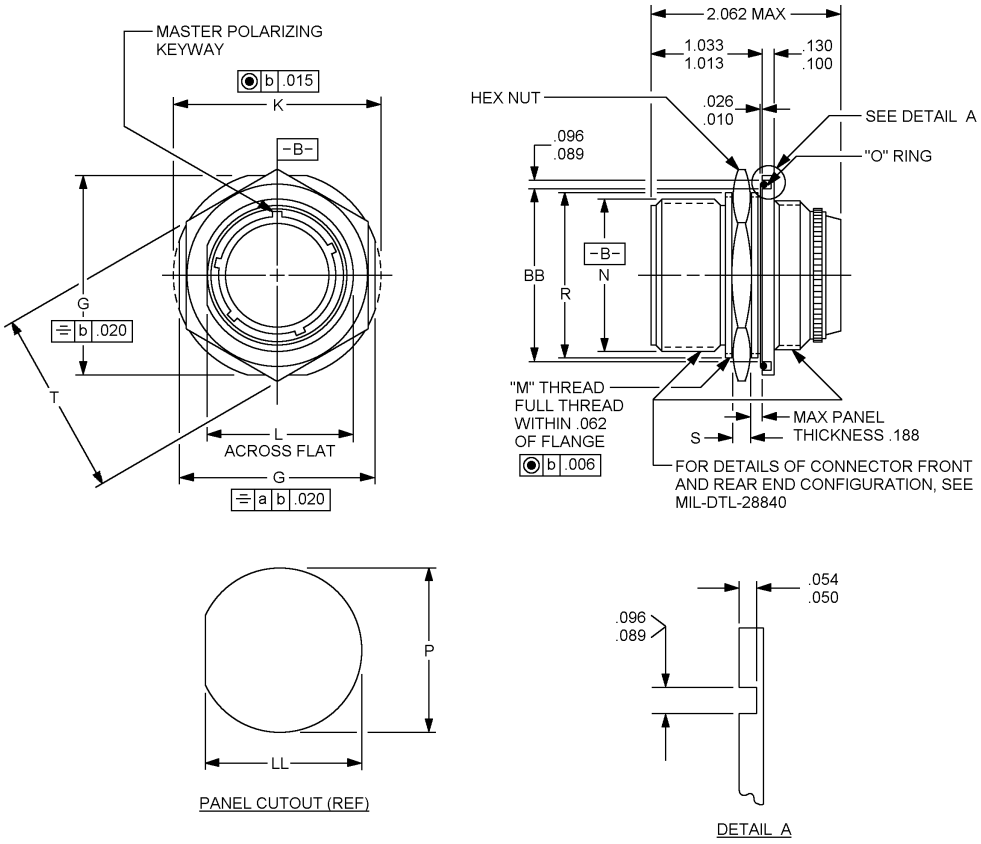


FIGURE 1. Dimensions and configurations.

AMSC N/A

FSC 5935

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T19:38Z
Check the source to verify that this is the current version before use.

Inches	mm	Inches	mm	Inches	mm	Inches	mm
.006	0.15	.050	1.27	.100	2.54	1.013	25.73
.010	0.25	.054	1.37	.125	3.18	1.033	26.24
.015	0.38	.062	1.57	.130	3.30	2.062	52.37
.020	0.51	.089	2.26	.188	4.75		
.026	0.66	.096	2.44	.189	4.80		

Shell size designator (see note 1)	G	K	L across flat	M thread class 2A
A (11)	1.274 (32.36) 1.254 (31.85)	1.368 (34.75) 1.348 (34.24)	.841 (21.36) .832 (21.13)	7/8-20 UNEF
B (13)	1.399 (35.53) 1.379 (35.03)	1.508 (38.30) 1.488 (37.80)	.966 (24.54) .957 (24.31)	1-20 UNEF
C (15)	1.587 (40.31) 1.567 (39.80)	1.681 (42.70) 1.661 (42.19)	1.153 (29.29) 1.144 (29.06)	1 3/16-18 UNEF
D (17)	1.649 (41.88) 1.629 (41.38)	1.743 (44.27) 1.723 (43.76)	1.216 (30.89) 1.207 (30.66)	1 1/4-18 UNEF
E (19)	1.837 (46.66) 1.817 (46.15)	1.931 (49.05) 1.911 (48.54)	1.403 (35.64) 1.394 (35.41)	1 7/16-18 UNEF
F (23)	2.024 (51.41) 2.004 (50.90)	2.118 (53.80) 2.098 (53.29)	1.591 (40.41) 1.582 (40.18)	1 5/8-18 UNEF
G (25)	2.149 (54.58) 2.129 (54.08)	2.243 (56.97) 2.223 (56.46)	1.716 (43.59) 1.707 (43.36)	1 3/4-18 UNS
H (29)	2.337 (59.36) 2.317 (58.85)	2.435 (61.85) 2.415 (61.34)	1.903 (48.34) 1.894 (48.11)	1 15/16-16 N
J (33)	2.524 (64.11) 2.504 (63.60)	2.618 (66.50) 2.598 (65.99)	2.091 (53.11) 2.082 (52.88)	2 1/8-16 UN

FIGURE 1. Dimensions and configurations - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T19:38Z
Check the source to verify that this is the current version before use.

Shell size designator (see note 1)	N max	P	R max	S hex nut thickness
A (11)	.750 (19.05)	.890 (22.61) .880 (22.35)	.875 (22.22)	.125
B (13)	.875 (22.22)	1.015 (24.78) 1.005 (25.78)	1.000 (25.40)	.125
C (15)	1.062 (26.97)	1.203 (30.56) 1.193 (30.30)	1.188 (30.18)	.125
D (17)	1.125 (28.58)	1.265 (32.13) 1.255 (31.88)	1.250 (31.75)	.125
E (19)	1.312 (33.32)	1.453 (36.91) 1.443 (36.65)	1.438 (36.52)	.125
F (23)	1.500 (38.10)	1.640 (41.66) 1.630 (41.40)	1.625 (41.28)	.125
G (25)	1.625 (41.28)	1.765 (44.83) 1.755 (44.58)	1.750 (44.45)	.189
H (29)	1.812 (46.02)	1.953 (49.61) 1.943 (49.35)	1.938 (49.22)	.189
J (33)	2.000 (50.80)	2.140 (54.36) 2.130 (54.10)	2.125 (53.98)	.189

FIGURE 1. Dimensions and configurations - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T19:38Z
Check the source to verify that this is the current version before use.

Shell size designator (see note 1)	T ± .017	BB	LL
A (11)	1.062	.979 (24.87) .969 (24.61)	.853 (21.67) .843 (21.41)
B (13)	1.188	1.104 (28.04) 1.094 (27.79)	.978 (24.84) .968 (24.59)
C (15)	1.375	1.292 (32.82) 1.282 (32.56)	1.165 (29.59) 1.155 (29.34)
D (17)	1.438	1.354 (34.39) 1.344 (34.14)	1.228 (31.19) 1.218 (30.94)
E (19)	1.625	1.542 (39.17) 1.532 (38.91)	1.415 (35.94) 1.405 (35.69)
F (23)	1.812	1.729 (43.92) 1.719 (43.66)	1.603 (40.72) 1.593 (40.46)
G (25)	2.000	1.854 (47.09) 1.844 (46.84)	1.728 (43.89) 1.718 (43.64)
H (29)	2.188	2.042 (51.87) 2.032 (51.61)	1.915 (48.64) 1.905 (48.39)
J (33)	2.375	2.229 (56.62) 2.219 (56.36)	2.103 (53.42) 2.093 (53.16)

NOTES:

1. Shell sizes are provided within parentheses for information and are not a part of the designator.
2. Dimensions are in inches.
3. Metric equivalents are given for information only.
4. Dimensions apply after plating.
5. Normal mating keyway position shown. For alternate keyway positions, see MIL-DTL-28840.

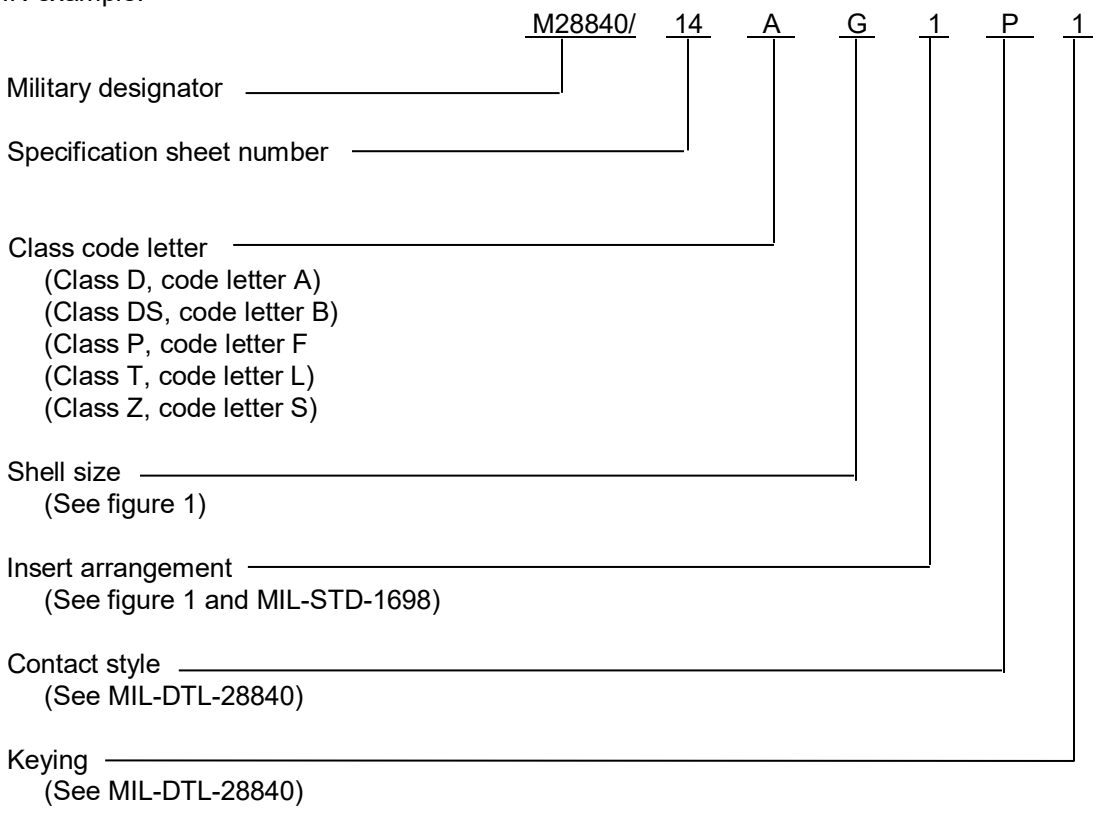
FIGURE 1. Dimensions and configurations - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T19:38Z
Check the source to verify that this is the current version before use.

REQUIREMENTS:

Dimensions and configurations: See figure 1.
Classes: D, DS, P, T and Z.
Insert arrangements: See the figure of MIL-STD-1698 corresponding to the desired shell size.
Contacts: Style P or S crimp type.
Shell polarization: 1 thru 6, key position.
Mating counterpart: Plug connectors specified in MIL-DTL-28840/16, MIL-DTL-28840/17, MIL-DTL-28840/18, MIL-DTL-28840/19, MIL-DTL-28840/26, MIL-DTL-28840/28, and MIL-DTL-28840/29.
Installation and removal tool: As specified in SAE-AS81969/33 and SAE-AS81969/34.
Mounting nuts shall have six equally spaced holes capable of withstanding a 50 pound pull in any direction when using the .030 inch minimum diameter wire specified in NASM33540. A backshell shall be used to meet moisture and water pressure requirements.
O-ring material: Fluorosilicone.
Marking: The cover shall be identified with the manufacturer's name or trademark and the Part or Identifying Number (PIN).

PIN example:



RESTRICTIONS:

Classes D and DS are inactive for NAVSEA new design, unless approved by NAVSEA.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T19:38Z
Check the source to verify that this is the current version before use.

Amendment notations. The margins of this specification are marked with vertical lines to indicate modifications generated by this amendment. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations.

Referenced documents. In addition to MIL-DTL-28840, this document references the following:

MIL-DTL-28840/16
MIL-DTL-28840/17
MIL-DTL-28840/18
MIL-DTL-28840/19
MIL-DTL-28840/26
MIL-DTL-28840/28
MIL-DTL-28840/29
MIL-STD-1698
NASM33540
SAE-AS81969/33
SAE-AS81969/34

CONCLUDING MATERIAL

Custodians:
Army - AT
Navy - EC
Air Force - 85
DLA - CC

Preparing activity:
DLA - CC

(Project 5935-2019-071)

Review activities:
Army – AV, MI
Navy – OS, SH

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T19:38Z
Check the source to verify that this is the current version before use.